

THORNEY ISLAND, United Kingdom

WMO: 038720

Lat: 50.8147N

Lon: 0.9225W

Elev: 3

StdP: 101.29

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.5	-1.3	-5.0	2.5	0.4	-3.8	2.8	0.8	15.2	9.6	14.0	9.7	2.9	30	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	25.3	19.0	23.5	18.3	22.0	17.4	20.0	23.9	19.0	22.3	18.2	21.0	4.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.4	21.5	17.6	12.6	20.5	16.9	12.1	19.7	57.1	24.1	53.8	22.4	51.2	21.1	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.9	10.5	DB	-4.9	28.0	1.4	2.1	-5.9	29.5	-6.7	30.8	-7.5	31.9	-8.5	33.4	(4)
(5)			WB	-4.9	21.8	1.3	1.9	-5.8	23.2	-6.6	24.3	-7.3	25.4	-8.3	26.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	5.7	5.7	7.2	9.5	12.5	15.5	17.6	17.5	15.2	12.2	8.8	6.2	(6)	
	DBStd	5.13	3.16	2.95	2.59	2.73	2.67	2.19	2.15	2.05	2.18	2.77	3.16	3.39	(7)	
	HDD10.0	588	136	121	90	40	7	0	0	0	0	12	59	124	(8)	
	HDD18.3	2656	392	353	344	266	182	91	40	40	95	189	287	377	(9)	
	CDD10.0	1012	2	1	4	24	84	163	236	232	157	81	22	6	(10)	
	CDD18.3	38	0	0	0	0	1	4	17	14	2	0	0	0	(11)	
	CDH23.3	173	0	0	0	0	4	24	83	57	5	0	0	0	(12)	
	CDH26.7	22	0	0	0	0	0	2	11	9	0	0	0	0	(13)	
(14) Wind	WSAvg	4.8	5.4	5.0	4.9	4.7	4.8	4.6	4.6	4.4	4.5	5.0	5.0	5.2	(14)	
(15) Precipitation	PrecAvg	857	100	63	63	56	51	57	50	60	65	99	94	100	(15)	
	PrecMax	1163	208	156	155	133	113	156	108	127	150	210	199	187	(16)	
	PrecMin	668	20	2	6	0	10	12	6	4	7	34	25	32	(17)	
	PrecStd	129	45	43	28	34	29	40	27	34	39	42	50	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	12.6	15.7	21.1	23.6	26.2	28.2	27.9	23.8	19.6	15.7	13.5	(19)	
		MCWB	11.2	10.5	11.4	14.5	17.0	19.4	19.6	20.1	18.5	16.0	14.5	12.3	(20)	
	2%	DB	11.6	11.4	13.5	17.5	21.1	23.3	25.6	24.7	21.6	17.9	14.7	12.7	(21)	
		MCWB	10.6	9.9	10.2	13.0	15.8	17.9	19.0	19.0	17.5	15.6	13.4	11.7	(22)	
	5%	DB	10.9	10.6	12.1	15.2	19.0	21.4	23.6	22.9	20.2	17.0	14.1	12.1	(23)	
		MCWB	9.9	9.3	9.4	11.5	14.9	16.9	18.5	18.3	16.5	15.1	12.8	11.1	(24)	
	10%	DB	10.2	9.9	11.1	13.6	17.1	19.8	21.9	21.5	19.1	16.2	13.3	11.3	(25)	
		MCWB	9.1	8.7	9.1	10.6	13.6	16.1	17.5	17.2	15.9	14.5	12.1	10.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.2	12.0	15.4	18.1	20.7	21.2	21.9	19.7	17.4	14.8	12.7	(27)	
		MCDB	12.0	11.9	14.9	19.1	22.1	24.4	25.6	25.5	22.3	18.8	15.3	13.1	(28)	
	2%	WB	10.8	10.3	11.0	13.4	16.4	18.6	19.8	20.0	18.3	16.3	13.8	12.0	(29)	
		MCDB	11.4	11.0	12.6	16.9	19.8	21.9	24.3	23.7	20.7	17.3	14.5	12.5	(30)	
	5%	WB	10.2	9.6	10.2	12.1	15.2	17.5	18.9	18.8	17.4	15.6	13.1	11.3	(31)	
		MCDB	10.8	10.3	11.5	14.4	18.2	20.6	22.7	21.6	19.4	16.6	13.9	11.9	(32)	
	10%	WB	9.4	8.9	9.5	11.2	14.1	16.5	18.0	18.0	16.7	15.0	12.3	10.4	(33)	
		MCDB	10.1	9.7	10.7	13.1	16.5	19.2	21.1	20.6	18.5	16.0	13.2	11.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.3	6.5	7.3	7.5	7.5	7.5	7.5	7.5	6.5	5.6	5.1	(35)	
		MCDBR	4.8	5.4	7.8	9.6	10.2	10.3	10.7	9.6	8.7	6.4	4.9	4.2	(36)	
	5% WB	MCWBR	4.6	4.5	5.5	6.1	6.2	6.2	5.8	5.3	5.3	4.6	4.3	4.2	(37)	
		MCWBR	4.7	4.9	6.5	8.1	9.3	9.0	9.5	8.2	7.6	5.6	4.6	4.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.334	0.350	0.387	0.405	0.408	0.406	0.399	0.393	0.379	0.364	0.337	0.325	(40)	
		taud	2.369	2.375	2.269	2.247	2.286	2.321	2.346	2.382	2.409	2.423	2.416	2.359	(41)	
	Ebn at Noon		681	774	805	834	848	852	852	839	809	748	675	628	(42)	
		Edh at Noon	65	83	109	124	125	122	117	108	94	78	62	57	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.83	1.54	2.74	4.35	5.25	5.80	5.51	4.61	3.42	1.93	1.01	0.65	(44)	
	RadStd		0.06	0.19	0.35	0.41	0.31	0.54	0.44	0.35	0.26	0.16	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page